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Improving cost–benefit analyses for health-considered climate mitigation policymaking

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There is growing discussion about enhancing climate policy efficiency by prioritizing health, with expectations for including health co-benefits in the next round of nationally determined contribution updates. Critical to this effort is the need to compare the benefits to the costs of mitigation. Here we synthesize the current cost-effectiveness of climate policies based on health-included cost–benefit analyses and identify key research challenges and opportunities for scaling up health-considered or even health-centred climate policies. Furthermore, we show factors essential to accelerating the development and implementation of mitigation policies, including providing tangible and policy-relevant health co-benefits, promoting interdisciplinary contributions and cross-sector policy engagement, conducting regional studies and improving inter-study comparability, and exploring health-considered optimized strategies.

Climate change mitigation can save lives and improve health through many pathways, including reducing air pollution, promoting active transport and shifting to healthier diets[1,2]. These more immediate health co-benefits could provide stronger incentives for more climate actions beyond avoiding unnecessary deaths and illnesses related to warming climate and extreme weather events. Recognizing the great potential to achieve climate, air quality[3] and health goals simultaneously, UN Climate Change Conferences have accelerated the integration of health into climate change agendas. The first-ever COP28 UAE Declaration on Climate and Health, signed by more than 150 Parties, urged for strengthening the development and implementation of mitigation policies that maximize the health co-benefits[4]. This unprecedented focus was extended through the 77th World Health Assembly[5] and culminated in the Baku COP Presidencies Continuity Coalition for Climate and Health[6]. With the next round of nationally determined contribution (NDC) renewals due by 2025, it is therefore critical to provide a comprehensive 'stocktake' on the costs, health co-benefits and the consequent cost-effectiveness of mitigation actions to better prioritize health gains while achieving NDC goals.

The bulk of the existing literature focuses predominantly on the health co-benefits of different mitigation actions (carbon pricing policy, energy transition, improvements in the agricultural and food sectors, and so on)[7,8] and health exposure pathways (ambient air pollution, physical activity, food intake and so on)[9–12] at the global[13–17] or regional scale[18–21]. Most studies provide evidence for large health gains from climate policy in different regions. Unfortunately, this scientific consensus has failed to trigger more ambitious mitigation actions due to a lack of tangible and policy-relevant economic interpretations of health co-benefits and cost–benefit analysis (CBA) studies[22–24], and to a lack of clear guidance on how to design a climate policy that takes health into consideration[25,26]. As a result, it is rare to see the incorporation of health co-benefits in real-world climate policies[27–29], let alone putting health at the centre of climate policymaking[21,30,31], as anticipated by the health community[5,32].

Therefore, to find out why the current literature fails to inform climate policies that incorporate health considerations, we first summarize the mitigation costs, health co-benefits and cost-effectiveness of climate actions from global and regional perspectives in current

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Table 1 | Global mitigation costs and health co-benefits of climate policies towards 2 °C or 1.5 °C targets

Research	Mitigation scenario (versus baseline)	Time	Carbon reduction (Gt)	Mitigation costs (trillion US$)	Health co-benefits (trillion US$)	BCR
Rauner et al.[35]	NDC	2050	10.7	1.8	1.9	1.06
	NDC—coal exit		26.4	2.7	2.9	1.07
	SSP226		52.0	3.2	3.5	1.09
Sampedro et al.[40]	SSP2—2 °C (all available)	2020–2050	610.1	17.0	37.2	2.19
	SSP2—2 °C (bioenergy limitation)		805.7	34.2	49.6	1.45
	SSP2—2 °C (low CCS)		596.9	18.4	36.7	1.99
	SSP2—2 °C (nuclear phase-out)		611.6	19.8	37.3	1.88
Markandya et al.[42]	SSP2—NDC	2020–2050	~450	7.5	11.8	1.57
	SSP2—2 °C (CER)		~700–950	22.1	54.1	2.45
	SSP2—2 °C (EPC)			28.3	51.8	1.83
	SSP2—2 °C (CAP)			41.6	67.5	1.62
	SSP2—1.5 °C (CER)		~950–1,200	40.6	69.3	1.70
	SSP2—1.5 °C (EPC)			39.7	69.0	1.74
	SSP2—1.5 °C (CAP)			56.1	78.4	1.40

The values shown in this table are the differences between mitigation scenarios and the no-climate-policy scenarios. The costs and benefits are presented in nominal US dollars. Rauner et al.[35] used 2005 US dollars, and Sampedro et al.[40] and Markandya et al.[42] used 2015 US dollars. The health outcomes assessed in Rauner et al.[35] are DALY, while those in Sampedro et al.[40] and Markandya et al.[42] are premature mortality. In Markandya et al.[42], the acronyms of scenarios are different equity allocation approaches. CER, constant emission ratios (maintains current emission ratios, preserves status quo); EPC, equal per capita (convergence towards equal annual emissions per person by 2040); CAP, capability (countries with high GDP per capita have low emissions allocations). The carbon reductions in Markandya et al.[42] are estimated approximately based on Fig. 1 and Table 2 in the original study which did not publish the exact values.

health-included CBA studies. Next, we provide an in-depth analysis of the challenges faced when designing a health-considered climate policy for real-world implementation. Finally, suggestions for the scientific community to promote health-considered or even health-centred climate mitigation targets, technology pathways and implementation strategies will be proposed, to better embed health co-benefits into climate mitigation policymaking.

Costs and health co-benefits of climate mitigation policy

CBA is an essential tool for evaluating the cost-effectiveness of mitigation actions, providing evidence to inform climate policymaking. Most existing CBA studies have explored the air-pollution-related health benefits of low-carbon energy transition. These studies evaluate the implementation costs and monetized health co-benefits across different mitigation policy scenarios and compare them with those in a baseline (that is, no-policy or business-as-usual) scenario. Notably, the health co-benefits discussed here refer to the local health benefits that accrue in areas where mitigation actions are implemented, while the direct climate-related health benefits, regarded as global public goods, are not included. The evaluation of health co-benefits entails the assessment of health outcomes, predominantly through calculating premature mortality, while several studies use the years of life lost[33,34] or the disability-adjusted life year as metrics[35,36]. These health outcomes are subsequently monetized into economic benefits, most of which are based on the value of a statistical life (VSL). The mitigation costs are usually presented as additional investments or expenditures on technologies, reduction in gross domestic product (GDP) or consumption loss[37,38].

Globally, air-pollution-related health co-benefits of climate policies usually outweigh the mitigation costs[33,35,39–46] (Table 1). If we define the cost-effectiveness of mitigation policy as the health co-benefits divided by the mitigation costs (benefit–cost ratio, BCR), the BCRs in these global studies usually range from 1.10 to 2.45, which means that an investment of US$1.00 in mitigation costs will yield a return of US$1.10–2.45 in health co-benefits globally[35,40,42].

It is worth noting that in the global studies with regional details, the cost-effectiveness of mitigation actions (that is, BCRs) varies considerably across regions. The difference of BCR can be as large as 40-fold[40]

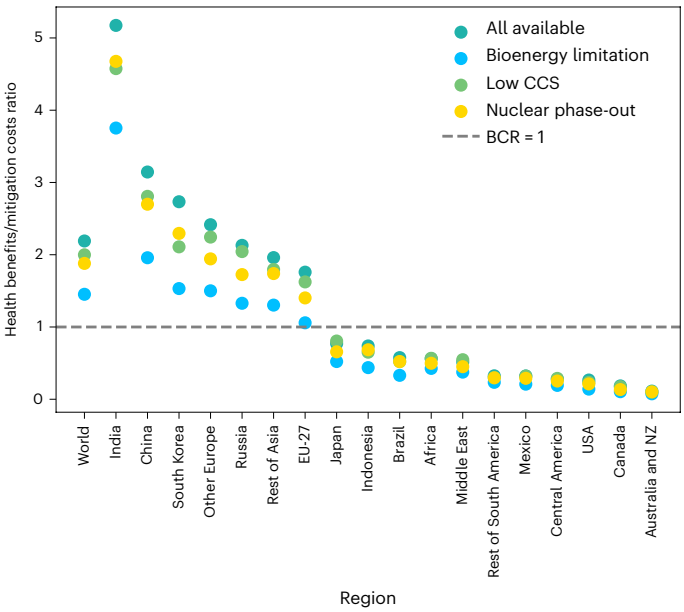

Fig. 1 | Global and regional cumulative health BCRs under different 2 °C scenarios from 2020 to 2050. Data are adopted from Sampedro et al.[40]. There are four technological pathways (shown by different colours) to achieving the 2 °C target. The grey dotted line, BCR = 1, is where the health benefits are equal to the mitigation costs of climate actions. CCS, carbon capture and storage.

(Fig. 1). There are two major reasons for substantial differences in regional health co-benefits. First, regions with higher levels of air pollution and population densities, such as India and China, have higher potential health co-benefits. Conversely, developed regions (Europe, the United States, Canada and Australia), where more stringent air pollution control policies have already been implemented, have lower health co-benefits. Therefore, Sampedro et al. concluded that the BCRs from 2020 to 2050 under the 2 °C constraint compared with the reference scenario were 3.75–5.17 in India and 1.95–3.15 in China, much higher than the BCRs of 1.06–1.76 in Europe and 0.14–0.27 in the

United States[40]. Second, there are substantial differences in VSL across countries because existing studies regionally extrapolate the VSL based on income levels (or GDP). Therefore, Africa, the Middle East and Latin America also show limited health co-benefits because of their low VSL, even with high air-pollution-related health impacts[47].

In addition, some pioneering studies have assessed the health co-benefits of national, subnational and even facility-level mitigation actions at higher spatial resolution; the majority of these attempts have been conducted in the United States[48–58], China[34,59–69], the United Kingdom[70,71] and Europe[36,72]. These regional studies reveal subregional differences in the cost-effectiveness of mitigation actions. For instance, the BCRs of energy transition in China are generally higher in the eastern regions, and some northwest provinces could potentially experience adverse health impacts resulting from climate policies[59,64,73–76]. Moreover, some regional studies have revealed inter-study conflicts, proving that mitigation actions in low-BCR regions in global studies, such as Europe[72], Brazil[77] and the United States[48,49,52,78–80], could be cost-effective. For example, the BCR from 2020 to 2030 is 0.72–1.62 for achieving 50% carbon reductions below the 2005 level in 2030 in the United States[48].

Existing studies also show that the scale of health co-benefits depends heavily on what specific climate actions are taken, leading to inevitably different CBA results in the same region. However, it is challenging to identify the cost-effectiveness of each technology or action because existing CBA studies present their results in a highly aggregated format, based on the original scenario design of technology portfolios. Nevertheless, there is a boom in the exploration of the health co-impacts of different low-carbon technologies or measures, such as negative emission technologies[34,81–83] and bioenergy[84,85], implying the potential to threaten health if mitigation policies are not well designed. Apart from air pollution, several CBA studies have discussed the health co-benefits from other exposure pathways, for instance, dietary changes[86,87] and physical activity[78,88,89]. There are trade-offs between these health co-benefits across exposure pathways. For example, switching to certain healthy dietary patterns, although generating huge weight- and nutrition-related health benefits, is likely to increase air-pollution-related health burdens in China due to higher ammonia emissions from fertilizing fruit and vegetables[90]. Promoting active transport could improve physical fitness[70], yet increase the risks of traffic injuries and fatalities[78] and harm from inhaling air pollutants[91]. These complex conclusions indicate that mitigation actions are not directly linked to positive health impacts.

Challenges for real-world implementation

Although previous studies have revealed substantial health co-benefits associated with the implementation of climate actions, these scientific findings have rarely been translated into health-considered climate policy and have failed to inform policymakers to take specific climate actions in real-world practices due to the following four challenges.

Lack of tangible and policy-relevant economic interpretations of health co-benefits

As described above, the VSL is frequently used to monetize environmental and health benefits. However, it is by no means a tangible and policy-relevant tool to trigger decision-making. The VSL is usually calculated from a self-reported willingness to pay for reducing fatal risk, which includes not only direct healthcare costs and indirect costs (for example, transportation and productivity loss) but also some additional non-use/intangible values (for example, the subjective painfulness of diseases). Hence, these VSL-based health co-benefits are not found in real economic accounts and are not comparable to realistic mitigation costs. Consequently, the conclusions from numerous CBA studies are not easy for policymakers to accept and implement in real-world decision-making processes[92].

In addition, the utility of existing studies to policymakers is limited by the considerable uncertainties over the VSL even across the same region[93,94]. For example, the VSLs used in CBA studies in China range from US$0.1 million to US$8.0 million (mostly in the US$0.3–1.0 million range)[62,95], and this wide range is sufficient to dominate the net benefits and introduce huge uncertainty in judging whether a climate policy is cost-effective[60–62,73,96]. Therefore, using the VSL to determine the monetized health co-benefits makes it difficult to provide persuasive conclusions to support climate policymaking.

Timescale and agent mismatches between mitigation costs and health co-benefits

The inherent timescale and agent mismatches between mitigation costs and health co-benefits (air-pollution-related benefits are discussed here) are critical realistic barriers to inform health-considered climate policymaking even if the mitigation costs are outweighed by the health benefits in total. Because low-carbon technologies deliver long-term environmental benefits over their lifetime, mitigation investments should be evaluated against the cumulative health co-benefits throughout their lifecycle in existing studies. However, this long-term cost–benefit equilibrium obscures critical near-term financial challenges during phases of technological deployment. When considering smaller time segments, the initial financial expenditures are considerable because low-carbon technologies usually have higher capital costs than traditional technologies, although in the long term, the health benefits gradually pay back the initial financial expenditure as the capital costs decline[46,97]. For example, the investment needed to develop renewable energy in China is much higher than the health co-benefits in 2030 although the net benefits are significant in 2050[74]. Consequently, although existing studies highlight the cost-effectiveness of climate policies, such findings often fail to impress policymakers with a clear trade-off between intergenerational and short-term considerations[40–42,46]. This financial challenge is particularly pronounced in developing areas in which the potential health co-benefits are also very large[98,99]. Therefore, few of these long-term net benefits, however great, are enough to impress policymakers to take immediate climate action.

In addition to timescale barriers, mismatches between the agents responsible for private mitigation costs and for social health co-benefits, essentially an externality problem in the field of environmental economics, also hinder the immediate implementation of climate action. Although air-pollution-related health co-benefits may be more persuasive for policymakers than climate-related benefits, the health co-benefits are still public goods and structural cost–benefit mismatches between economic agents persist. Specifically, it is often carbon-intensive industries that are bearing the financial burden of carbon reductions. From their perspective, especially for companies that lack environmental, social and governance awareness, the health co-benefits borne by surrounding residents are not usually considered in their economic thinking. Furthermore, these agent mismatches pose governance challenges for cross-sector issues due to the separate decision-making processes implemented by government ministries or agencies with different priorities. For instance, companies are regulated by economic, energy and industry ministries, which inadequately incorporate health into their policies, whereas public health benefits are the responsibility of environmental, health and social ministries. Therefore, the highly cost-effective mitigation actions proven in existing studies remain difficult to facilitate in real-world practice without additional mechanisms to redirect health co-benefits to cost-bearers. In summary, the inherent timescale and agent mismatches between mitigation costs and health co-benefits explain the difficulty in implementing climate policy, despite its general cost-effectiveness.

Lack of region-specific health co-benefits for climate policymaking

Despite the broad consensus on the health co-benefits of carbon reduction at the global level, the cost-effectiveness implications of national or subnational climate policies remain unclear. The low

spatial resolution of global studies means they are unable to support region-specific policymaking. Furthermore, the diverse findings of different CBA studies—often using different parameters, assumptions and methodologies—pose challenges in generating comprehensive and solid conclusions for certain regions. In particular, few CBA studies in Africa and the Middle East include national or regional details[100,101]. These are regions in which carbon emissions are increasing and studies are urgently needed to support climate actions there. Consequently, the global awareness of cost-effective climate policy is impeded from being translated into region-specific actions due to the lack of regional studies with higher spatial resolutions, especially in the Global South.

Lack of clear guidance on health-considered climate policymaking

As previously discussed, health co-benefits depend heavily on what specific climate actions are taken. Therefore, to effectively promote health-considered climate policy, it is imperative to provide clear guidance on choosing technology pathways, rather than merely demonstrating the potential for carbon reductions of aggregated technology portfolios. However, existing studies struggle to satisfy the need for complex decision-making in real-world implementations because they are comparing mitigation costs and health co-benefits in limited scenarios with poor technology details and single health exposure pathways. Although recent studies have expanded to assess broader mitigation technology portfolios across multiple sectors and health-exposure pathways, they do not lead to more decisive policy implications because they provide complex (and sometimes conflicting) conclusions. These findings highlight that differentiated impacts of low-carbon technologies and exposure pathways should be carefully considered to avoid any potential negative impacts on health.

Furthermore, existing CBA studies fail to encompass the substantial health co-benefits accruing from multiple exposure pathways such as improving indoor air quality and thermal comfort by using clean household energy[70,71,102–105], and reducing occupational exposures associated with fossil fuel extraction[106]. Furthermore, although some current CBA studies have investigated air-pollution-related respiratory and cardiovascular health outcomes, newly discovered pathophysiological pathways, such as diabetes, obesity and mental health outcomes, are generally not included[107,108]. Policymakers need to know the CBA of more low-carbon technologies[40] and health co-impacts through different health exposure pathways[102,103,109,110], and further identify the priority issues and explore better climate policies to deliver more health co-benefits or net benefits. In other words, researchers need to provide clear guidance on decision-making by optimizing health-considered mitigation pathways.

Avenues to support health-considered climate mitigation policymaking

Despite the limitations of existing studies, the scientific community can provide clear policy implications and accelerate health-considered or even health-centred mitigation actions in the following four unique ways.

Provide tangible and policy-relevant economic interpretations of health co-benefits

Future studies need to consider alternative ways to assess the monetized health co-benefits rather than using VSL-based approaches. For example, some studies have used the cost-of-illness method[60,77,86,111,112], which incorporates data on actual regional healthcare expenditures and calculates how mitigation actions can help reduce the burden on the healthcare system (or health insurance). A few studies have further assessed direct and indirect health-related economic impacts, such as work days lost and reduced labour productivity[41,74,113,114]. Economic impacts are of great concern to policymakers because governments do care about the numbers and need to pay for some of these losses. By understanding the tangible and policy-relevant health co-benefits of carbon reductions, policymakers can directly make better-informed decisions.

Promote interdisciplinary contributions and cross-sector policy engagement

To address the timescale and agent cost–benefit mismatches, discussions on new policies or decision-making frameworks with interdisciplinary contributions are required. By combining knowledge of environmental health, management and finance, studies could explore effective mechanisms to redirect health co-benefits to cost-bearers, to help incorporate long-term health impacts into traditional carbon-pricing policies[115,116]. Studies could expand assessments on policies targeting individual efforts without agent mismatch problems, such as household energy transition and demand-side mitigation actions (for example, promoting active transport and changes in diet), by integrating interdisciplinary methods. Apart from the total health benefits used in traditional CBA studies, researchers should quantify the feasibility and acceptability of mitigation actions[117,118], and further navigate the complex decision-making framework by incorporating more realistic factors[119–121]. To this end, it is essential that economists, political scientists and sociologists join the climate and health community and provide more specific guidance on complex policymaking from different disciplines.

More engagement between researchers and cross-sector policymakers is essential to address climate-health governance challenges[122]. Studies could provide targeted conclusions and raise awareness of cross-sector issues among policymakers from different government ministries or agencies (for example, climate, energy, economy, employment and health). It is important to foster greater policy willingness by highlighting the core benefits of health-considered mitigation actions in the priority lists from different sectoral policymakers (such as achieving air-quality guideline levels, safeguarding energy security and promoting economic development)[123]. More importantly, to bridge the separate decision-making processes, the involvement of higher-level decision-makers (for example, the state council, a leading group for climate actions) is crucial for establishing interdepartmental collaboration frameworks. More effective leadership and the consequent coordinated policies will be able not only to directly help to systematically address the timescale and agent mismatches between mitigation costs and health co-benefits but also balance other priority issues.

Conduct regional studies and improve inter-study comparability

To address critical knowledge gaps in regional health co-benefits, it is necessary to conduct further national, subnational or even city-level studies that fully consider local conditions to facilitate real-world implementation of climate actions. The geographic coverage of future studies needs to be expanded substantially and include higher spatial resolutions, particularly across the developing Global South, for which detailed analyses are lacking[17]. To achieve this, essential datasets and low-cost health-impact assessment tools need to be developed and integrated into policy assessment frameworks[24,124].

To derive comprehensive and solid conclusions for region-specific climate policymaking, it is necessary to synthesize disparate CBA studies. Therefore, it is critical to improve inter-study comparability by following the same modelling framework and data-disclosure protocols for future studies[125]. Details of mitigation actions, health exposure pathways and health outcomes should be clearly stated. For example, one commendable practice is the Pathfinder Initiative, which constructs a database on the health impacts of mitigation actions caused by different health exposure pathways, sectors and regions based on 57 original studies, with a uniform assessment framework and the same health outcomes (years of life lost)[15]. Inspired by this, it is recommended that the health impacts and benefits across scenarios be systematically collected in the scenario

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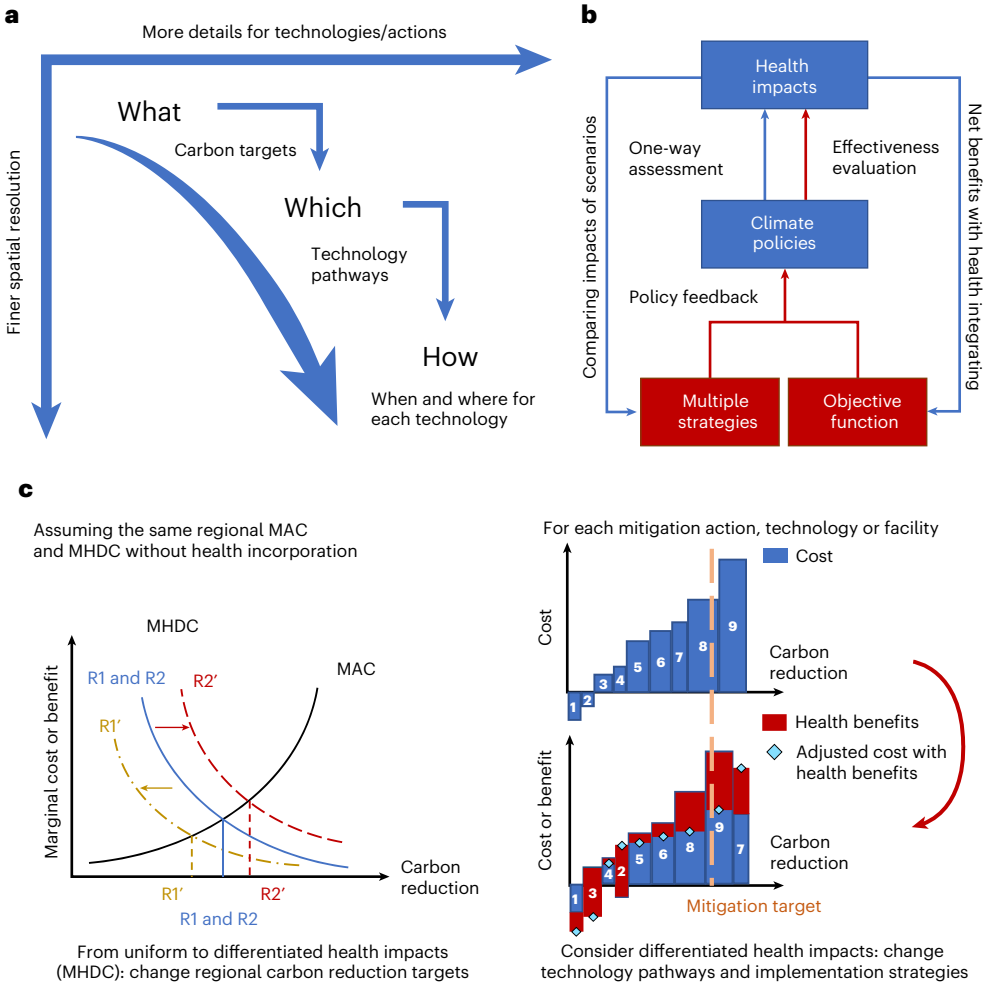

Fig. 2 | Conceptual framework for optimizing health-considered mitigation pathways. a, The relationship between three different research questions as described in the text. **b**, An additional method (in red) to optimize climate policy with health co-benefits compared with current one-way assessment studies. **c**, The mechanism of how the differentiated health benefits (in red) impact climate policymaking. Left: the modified marginal damage curves when including different health benefits by regions. Right: the modified MAC when including different health benefits by mitigation actions, technologies or facilities. MAC, marginal abatement cost; MHDC, marginal health damage cost.

database and compared with mitigation costs to more effectively embed health co-benefits into regional climate policies.

Explore the optimized mitigation pathways with health considerations

Although the concept of incorporating health co-benefits into climate policy has been proposed for a long time[126], more explorations on optimizing mitigation pathways while considering health are needed. Such research could facilitate a shift in decision-making for climate policies from traditional cost minimization to net-benefit maximization. A summary of cutting-edge studies suggests that additional research efforts should be encouraged to focus on the following three main policy-oriented questions (Fig. 2a). First, 'what' is the optimized allocation of carbon reduction targets or intensities among regions and sectors considering differences in ease of mitigation and the level of health co-benefits? Future studies need to incorporate health co-benefits into the objective functions (Fig. 2b) in models, offsetting (part of) the mitigation costs[33,47,127,128] (Fig. 2c, left).

Second, 'which' is the optimized technology pathway producing maximum health co-benefits in a specific region or sector under a given carbon reduction target? Investigations are needed to find the best technology pathways yielding more health or net benefits through scenario comparisons and objective function modification (Fig. 2b). To determine the optimum low-carbon technology pathways offering greater health improvements, researchers need to design extensive ensembles of scenarios with different technology mixes and compare their mitigation costs and health co-benefits[34,40,56]. In addition, future studies could reoptimize the health-considered technology pathways by internalizing the health damage costs of different actions or technologies into the model)[36,129–134] (Fig. 2c, right).

Third, 'how' should a given low-carbon technology pathway be implemented at higher spatial resolution (that is, using regionally tailored or plant-by-plant/facility-by-facility strategies), taking into account the regional and technological differences in its health impacts? The answer to this question requires considering the rankings of the carbon-intensive technologies to be retired or retrofitted and the low-carbon technologies to be deployed by these regions or plants/facilities. To provide a scientific basis for policymaking, researchers need to conduct finer-grained assessments of the carbon reduction and health co-benefits of mitigation actions and separate their contributions by region, action[135,136] or even plant/facility[137–140]. Based on these heterogeneities, optimized implementation strategies incorporating health co-impacts by technologies or facilities are needed (Fig. 2c, right), in contrast to traditional least-cost strategies. Several pioneering case studies have further suggested mitigation strategies with maximum health co-benefits being achieved by phasing out coal-fired

power stations[116,141–146], retiring fossil-fuel blast furnaces[147] and replacing fossil-fuel vehicles[148,149], and future studies should expand to consider more regions and technologies. By targeting the key policy issues discussed above, researchers can guide realistic decision-making and improve the efficiency of climate policy by maximizing the net benefits.

Conclusion

Climate change mitigation has great potential to generate health co-benefits and cost savings. The COP28 UAE Declaration on Climate and Health led to the expectation that leveraging mitigation actions with substantial health co-benefits and strengthening the implementation of efficient climate policies with maximum health gains would be encouraged in the next round of NDC renewals, due by 2025. To better embed health co-benefits of mitigation actions into climate policy, a 'stocktake' on the costs, health benefits and the consequent cost-effectiveness of mitigation actions from global and regional perspectives in current health-included CBA studies has been summarized here.

However, current studies lack the detail that would enable policymakers to take specific climate actions immediately, despite increasing evidence of the health co-benefits of various mitigation actions. The research challenges in designing a health-considered climate policy for real-world implementation are (1) a lack of tangible and policy-relevant economic interpretations of health impacts, (2) timescale and agent mismatches in terms of costs and benefits, (3) a lack of regional studies and (4) a lack of clear guidance on health-considered climate policymaking.

In summary, the scientific community is encouraged (1) to provide tangible and policy-relevant health co-benefits in real economic terms, (2) to promote interdisciplinary contributions and cross-sector policy engagement to address timescale and agent cost–benefit mismatches and the consequent governance challenges, (3) to conduct more regional studies and improve inter-study comparability to derive comprehensive and solid conclusions for region-specific climate policymaking, and (4) to explore the optimized health-considered climate target allocations, technology pathways and implementation strategies. These avenues for future researchers can ensure better engagement with policymakers and effectively accelerate the implementation of mitigation actions with maximum health co-benefits.

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Author contributions

J.S., C.Z., W.C. and C.W. conceived and designed the paper. J.S., S.Z. and M.Z. contributed material. J.S. and W.C. drafted the paper with input from all co-authors. All authors read and approved the final version of the paper.

Competing interests

The authors declare no competing interests.

Additional information

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